Visakh Padmanabhan

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LinkedIn | GitHub | Portfolio | Medium

Profile Summary

AI Engineer with three years in Germany building and shipping production **Large Language Model (LLM)** assistants and **agentic** workflows at fast-paced startups, taking early ideas from prototype to product. Pairs that delivery record with award-winning research: **Best Outgoing Student** and a **JURIX 2024 Best Student Paper** on hybrid legal retrieval. Works across **Retrieval-Augmented Generation (RAG)**, agentic retrieval, tool calling, and **LLM-as-judge** evaluation, measuring quality and cost rather than assuming them. **EU Blue Card** holder (full work authorization), open to relocation.

Technical Skills

- **Programming: Python** (OOP, NumPy, Pandas, Scikit-learn), **SQL**
- **AI/ML & LLM Frameworks: LangChain**, **LangGraph**, **LangSmith**, Sentence-Transformers, **RAG**, **agentic retrieval**, **tool calling**, **LLM-as-judge evaluation**, **context engineering**, **Knowledge Graphs**
- **Software Development & APIs: REST APIs**, **FastAPI**, Flask, Microservices
- **Databases & Vector Stores: PostgreSQL**, **pgvector**, Redis
- **Cloud Platforms & DevOps: AWS**, **Azure** (Azure AI), **Docker**, GitHub Actions, **CI/CD**, MLOps
- **Tools & Workflow:** Git, JIRA, Agile/Scrum
- **Languages:** English (Fluent, C1), German (Basic, A2)

Professional Experience

AI & Software Engineer *January 2026 – Present*
MaindTec GmbH, Ingolstadt, Germany

- Shipped production **Retrieval-Augmented Generation (RAG)** features end to end on MaiQ (an AI product-management tool), from **prototype to product**, spanning document processing, retrieval, and conversational interfaces.
- Extended **agent-building** from engineers to end users by shipping the **Skill Store** in MaiQ, where anyone can author, share, and run **their own agents** directly in chat through conversational skill authoring.
- Powered **semantic retrieval at scale** by building a file-ingestion service that parses and chunks PDFs into pgvector, using a **CUDA**-accelerated design and asynchronous Redis Streams workers to process **very large files and many files in parallel**.
- **Cut inference costs by 37%** with only a 2% drop in retrieval quality, running **11+ chunking experiments** under an **LLM-as-judge** evaluation framework and productionizing the best-performing strategy.
- **Improved document relevance** for end users with **hybrid search** (BM25 plus embedding-based retrieval) and **agentic retrieval** via LLM tool calling in LangChain.
- **Reduced token usage** across long conversation sessions through **context engineering**, pruning chat history and abstracting tool calls with agentic middlewares in LangChain.

Junior Data Scientist *January 2025 – August 2025*
Vay Technology, Berlin, Germany

- **Reduced ETA error by 25%** by statistically evaluating routing performance, working directly with operations stakeholders.
- **Enabled early detection of operational issues** by building **dbt models** that monitored rebalancer missions and gave stakeholders clear, reliable performance reports.
- **Cut decision-making latency** in production by integrating a **gRPC API** into vehicle rebalancing.

Data Scientist, Working Student *May 2022 – December 2024*
Einbliq.io, Munich, Germany

- **Owned the Detailed Device Data Service end to end**, building from scratch a real-time serverless Azure Functions pipeline that reached **99% coverage** of streaming device data.
- **Estimated power emissions and carbon footprint** from that device data using **machine learning** with **confidence scores**, delivering reliable device-level insights.
- **Gave the team continuous visibility** into streaming sessions and device capabilities by building **Grafana** dashboards over the Detailed Device Data Service.

Systems Engineer *October 2019 – August 2021*
Infosys R&D Ltd, Thiruvananthapuram, India

- Processed **unstructured text data** using **NLTK** and **spaCy** to identify and redact sensitive information via **named entity recognition** and text preprocessing.

Education

M.S. in Data and Knowledge Engineering *October 2021 – March 2025*
Otto von Guericke University, Magdeburg, Germany **Grade: 1.4**
B.Tech in Computer Science and Engineering *July 2015 – August 2019*
Mar Baselios College of Engineering and Technology, Thiruvananthapuram, India **Grade: 1.9**

Projects and Publications

- **Hybrid Legal Norm Retrieval** – Leveraging Knowledge Graphs and Sentence-BERT for retrieving Civil Code Articles (COLIEE 2024 Task 3). *Awarded Best Student Paper, JURIX 2024.* [**Link**]
  - Designed a hybrid retrieval pipeline combining **knowledge graph traversal** with **Sentence-BERT** embeddings to retrieve relevant legal norms from the Japanese Civil Code, achieving competitive results on the COLIEE 2024 benchmark.
- **LUMI: Legal Understanding and Matching through Interactive Highlighting** – Improved legal text explainability by highlighting semantically aligned spans. [**Link**]
- **Expert Agent Guided Learning** – Chatbot for guided learning using knowledge graphs built from OCR-extracted textbook content. [**Link**]
- **Slide Recommendation System** – OCR-based keyword extraction and ranking system to enhance student learning from slides. [**Link**]
- **MindMate** – Conversational AI Agent prototype using **LangGraph**, **LangChain**, and OpenAI API to deliver empathetic, context-aware dialogue with personalized coping strategies. [**Personal Project Link**]

Achievements

- **Best Outgoing Student Award (M.S.)** – Otto von Guericke University, Magdeburg
- **Best Student Paper Award, JURIX 2024** – Hybrid Legal Norm Retrieval: Leveraging **Knowledge Graphs** and **Sentence-BERT** for Retrieving Civil Code Articles (COLIEE 2024 Task 3)
- **All Round Performance Award** – Best Outgoing Student (2015–2019)